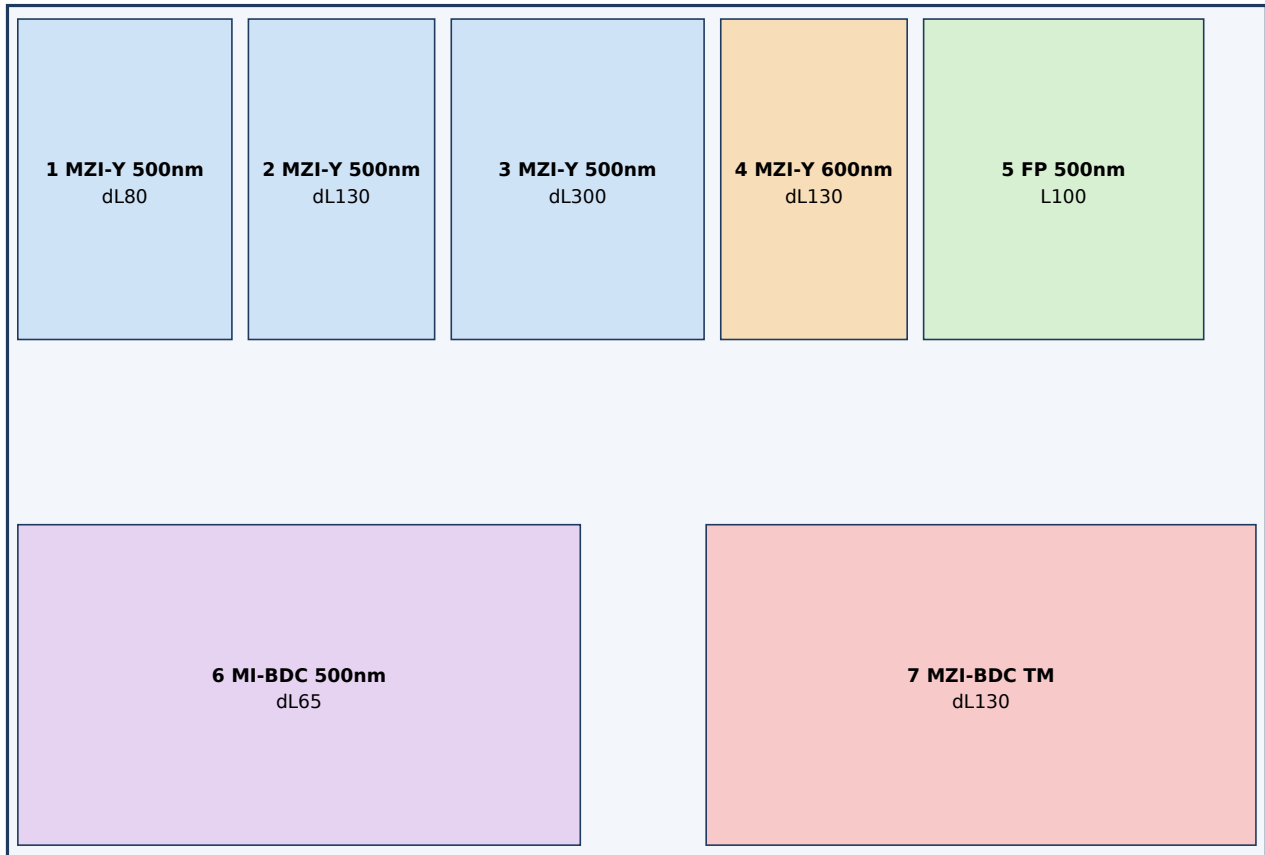


EBeam_AngelosPappas: Die Device Guide

openEBL 2026-10 submission | 605 × 410 μm die | 7 optical test structures

What this die is for

Every structure on this chip is an **interferometer or resonator whose fringe spacing (FSR) depends on the waveguide group index n_g** . Measuring the transmission spectrum around 1550 nm and reading off the fringe spacing gives n_g directly. Devices differ in waveguide width (500 vs 600 nm), polarization (TE vs TM), splitter type (Y-branch vs broadband directional coupler) and path-length difference, so n_g can be compared across those variables and cross-checked between structures.



Approximate footprint of each device (to scale). Number = row in the table below. Grating couplers sit at the left edge of each block (right edge for device 7), 127 μm apart (fiber-array pitch).

Device list

#	Label (after opt_in_..._AngelosPappas_)	Structure	What it measures
1	MZI_Y_TE500_dL80	Mach-Zehnder, Y-branch splitters, 500 nm TE, $\Delta L = 80 \mu\text{m}$	n_g from FSR (largest FSR)
2	MZI_Y_TE500_dL130	Same, $\Delta L = 130 \mu\text{m}$	n_g ; second ΔL as cross-check
3	MZI_Y_TE500_dL300	Same, $\Delta L = 300 \mu\text{m}$	n_g with finest fringes (best precision)
4	MZI_Y_TE600_dL130	Same as #2 but 600 nm wide waveguide (500↔600 nm tapers at the arms)	n_g vs. width: compare directly with #2
5	FP_TE500_L100	Fabry-Pérot: two loop mirrors (directional coupler + loop) around a $\sim 100 \mu\text{m}$, 500 nm cavity	n_g from cavity resonances; independent method

#	Label (after opt_in_..._AngelosPappas_)	Structure	What it measures
6	MI_BDC_TE500_dL65	Michelson: broadband directional coupler (BDC) + 2 arms ending in Y-branch loop mirrors, $\Delta L = 65 \mu\text{m}$	n_g with a BDC; light returns to the input port
7	MZI_BDC_TM500_dL130 (TM)	Mach-Zehnder, 2 TM broadband couplers, TM gratings, unused ports terminated, $\Delta L = 130 \mu\text{m}$	TM n_g ; compare with TE (#2, #6)

Naming: **MZI** Mach-Zehnder, **MI** Michelson, **FP** Fabry-Pérot, **Y** Y-branch splitter, **BDC** broadband directional coupler, **TE/TM** polarization, number after TE/TM = waveguide width in nm, **dL** = path-length difference in μm , **L** = cavity length in μm .

How to extract the group index

1. Measure the transmission spectrum of the device (labels are formatted for the automated opt_in measurement).
2. Remove the grating-coupler envelope (e.g. fit a low-order polynomial to the spectrum and subtract it), leaving only the fringes.
3. Find neighbouring peaks (or nulls) and take their spacing, the FSR, at each wavelength.
4. Compute n_g with the formulas below; fit $n_g(\lambda)$ around 1550 nm if you want the dispersion.

Structure	Formula	Expected FSR @1550 nm*
MZI (#1-4, #7)	$n_g = \lambda^2 / (\text{FSR} \cdot \Delta L)$	dL80: $\sim 7.1 \text{ nm}$; dL130: $\sim 4.4 \text{ nm}$; dL300: $\sim 1.9 \text{ nm}$
Fabry-Pérot (#5)	$n_g = \lambda^2 / (2 \cdot L \cdot \text{FSR})$	$\sim 2.9 \text{ nm}$
Michelson (#6)	$n_g = \lambda^2 / (2 \cdot \text{FSR} \cdot \Delta L_{\text{arm}})$ (light passes each arm twice)	check against the design

*Rough guide assuming $n_g \approx 4.2$ for 500 nm TE strip waveguides; real values will differ.

Notes and caveats

- **No reference structure:** the die has no straight loop-back waveguide, so the grating-coupler response has to be removed from each spectrum itself (step 2).
- **Bends and tapers count:** the nominal ΔL is the label value; the arms also differ in bend and taper content. Use the as-designed ΔL , and expect some systematic error in n_g from bend and coupler dispersion.
- **Quick sanity checks:** #2 vs #4 isolates the effect of width; #2 vs #7 isolates TE vs TM; #1/#2/#3 should all give the same n_g , and any spread is your measurement uncertainty.
- **Fabry-Pérot (#5):** the effective cavity length depends on how deep light penetrates the loop mirrors, so treat n_g from this device as a cross-check, not the primary result.
- **Michelson (#6) ΔL :** confirm whether the $65 \mu\text{m}$ is the one-way arm difference before applying the factor 2.